CITIZEN FRAUD SHIELD

AI-Powered Detection for Digital Arrest Scams

Detailed Project Report

ET AI Hackathon 2026

Problem Statement 6: AI for Digital Public Safety

Team Task Force 141

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Repository: github.com/Manan-mp/ET-hackathon-prototype

1. Executive Summary

Citizen Fraud Shield is an AI-powered detection tool built to counter "digital arrest" scams — a rapidly growing category of fraud in which criminals impersonate CBI, ED, Customs, or police officers over video and voice calls, psychologically isolate victims, and extort money under the threat of a fabricated arrest. The tool allows a citizen to paste a call or message transcript and receive, within seconds, a structured risk verdict, an explanation of the specific red flags detected, and plain-language guidance on what to do next.

The prototype was built for the ET AI Hackathon 2026 under Problem Statement 6: AI for Digital Public Safety, and is fully functional end to end — a working FastAPI backend, a Gemini-powered classifier, a persistent session log, and a responsive frontend.

2. Problem Statement & Context

India recorded 1.14 million cybercrime complaints in 2023, a 60% increase over the previous year. Digital arrest scams alone accounted for over ₹1,776 crore in losses in just the first nine months of 2024, per Ministry of Home Affairs data. These are not opportunistic crimes — they are coordinated operations that follow a consistent, exploitable pattern.

The Recurring Pattern

- Fraudster establishes a fake official identity (CBI, ED, Customs, police, RBI, TRAI)
- Victim is threatened with immediate legal action or a "digital arrest"
- Victim is instructed to stay on a continuous video call and isolated from family or advice
- Victim is pressured into transferring money, sharing OTPs, or making "verification" payments

The Gap

Existing response mechanisms are entirely reactive: a victim can only file a complaint after money has already been lost. There is no tool available at the point of contact — during the call itself — that can help a citizen recognize the scam in progress and act before harm occurs. Citizen Fraud Shield is designed to close exactly that gap.

3. Proposed Solution

Citizen Fraud Shield gives any citizen a simple, three-step way to evaluate a suspicious call or message:

Step 1 Analyze —

The citizen pastes the transcript of a call or message into the app. No account, installation, or technical knowledge is required.

Step 2 Detect —

An AI classifier, trained via a carefully constructed prompt on known digital-arrest scam patterns, evaluates the transcript and identifies specific, evidence-backed red flags rather than returning an opaque score.

Step 3 Protect —

The citizen receives a clear verdict (SAFE / SUSPICIOUS / HIGH-RISK SCAM), a confidence score, and a plain-language recommended action — available in Hindi, Tamil, Telugu, and Bengali in addition to English.

4. System Architecture

The system is intentionally lean: every hop between the citizen and an answer serves a clear purpose, which keeps latency low and the codebase easy to reason about.

Stage	Component	Role
1	Citizen	Pastes a call or message transcript into the app
2	Frontend	Single-page HTML/JS UI with sample scenario buttons
3	FastAPI backend	Exposes /api/analyze and /api/sessions routes
4	Gemini 2.5 Flash	Few-shot classifier returns a strict-JSON verdict
5	SQLite log	Persists verdict, confidence, and timestamp per session
6	Dashboard	Renders the color-coded result and recent-session history

Response loop: Gemini returns a strict-JSON verdict with confidence and red flags → the backend logs the session to SQLite → the result renders as a color-coded card on the frontend → recent sessions populate the live detections dashboard.

5. Key Features

- Scam Risk Analyzer — color-coded verdict, confidence score, and evidence-backed red flag cards
- Guided Reporting Panel — static, factual steps: the 1930 cybercrime helpline, cybercrime.gov.in, and nearest police station
- Recent Detections Dashboard — a live feed of the last 20 analyzed sessions with verdict badges
- Multilingual Advisory — recommended actions translated into Hindi, Tamil, Telugu, and Bengali
- Sample Scenarios — four pre-written transcripts (obvious scam, subtle scam, genuine call, medium risk) for a reliable, repeatable demo
- Session Logging — every analysis is persisted to SQLite with a timestamp for an auditable history

6. Technology Stack

- Backend: FastAPI on Python 3.12, with Pydantic v2 for request and response schemas
- LLM: Google Gemini 2.5 Flash, used as a few-shot scam-pattern classifier
- Database: SQLite for lightweight, zero-configuration session logging
- Frontend: A single-page HTML interface using vanilla JavaScript and Tailwind CSS via CDN

7. API Reference

Method	Path	Description

POST	/api/analyze	Analyze a transcript for scam patterns
GET	/api/sessions	List recent analyzed sessions
GET	/api/samples	Get pre-written demo scenarios
POST	/api/translate	Translate advisory text
GET	/	Serve the frontend

8. Implementation Methodology

Prompt Design

The Gemini classifier is guided by a system prompt that anchors it with few-shot examples of both a genuine digital-arrest scam transcript and a genuine benign call. The model is required to justify every red flag against evidence actually present in the transcript, and is instructed to be conservative: an ambiguous transcript is classified as SUSPICIOUS rather than confidently SAFE or HIGH-RISK.

Structured Output

The model is constrained to return strict JSON — verdict, confidence, a list of red flags with explanations, and a recommended action — which the backend parses directly into the response schema. This avoids brittle text-parsing and keeps the frontend rendering reliable.

Reliability for Demonstration

Four fixed sample transcripts are bundled with the application so that a live demonstration does not depend on typing a transcript in real time or on the variability of a freshly composed prompt.

9. Business Impact & Scalability

Impact

- Shifts intervention from post-loss investigation to in-the-moment prevention
- Directly targets the ₹1,776 crore annual loss pool attributed to digital arrest scams alone
- Every logged session strengthens a shared, growing fraud-pattern intelligence base

Scalability

- The FastAPI service is stateless and can scale horizontally behind a load balancer
- SQLite can be swapped for PostgreSQL to support multi-region, multi-agency deployment
- Natural distribution channels include the WhatsApp Business API, IVR systems, and bank/UPI apps

Roadmap

- Near term: live audio transcription for real phone calls, not just pasted text
- Mid term: a deepfake and AI-generated voice detection layer alongside the text classifier
- Long term: direct integration with NCRB reporting workflows and telecom carrier alerts

10. Team & Repository

This project was built by Team Task Force 141 for the ET AI Hackathon 2026.

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